

LINGUISTIC & DIALECTAL ROBUSTNESS

Arabic Fake News Detection – The Dialect Gap

Student: Mohamed Shreif Abdelsattar | ID: 2405467 | Supervised by Prof. Mohamed M.Abbassy

— MODULE 01 —

THE PROBLEM

Arabic Diglossia – why dialect breaks MSA-trained models

\$ Why Dialectal Robustness Matters

— The Core Challenge

- ▶ Arabic has a 'High' form (MSA) and many 'Low' forms (Egyptian, Levantine, Gulf...)

— The real threat:

- ▶ Propagandists use dialects to establish false intimacy – or to bypass MSA filters

- ▶ Model trained on formal news may read a sarcastic Egyptian tweet as 'Credible'

- ▶ # # It doesn't recognize informal negation like مش instead of لا

- ▶ → The Dialect Gap = drop in performance from MSA to real-world social media

KEY METRICS

What we measure to quantify dialect robustness

\$ The 3 Linguistic Metrics

METRIC

- ▶ Dialect Robustness Index
- ▶ Morphological Sensitivity
- ▶ Style-Veracity Correlation
- ▶ Prediction Flip Rate

WHAT IT ANSWERS

- ▶ Is the model equally fair across all Arabic regions?
 - ▶ Does accuracy drop when dialect prefixes/suffixes appear?
 - ▶ Does the model label informal text as Fake regardless of facts?
 - ▶ How often does dialect alone change the model's decision?

EXPERIMENT SETUP

Three versions of the same data – one model under pressure

\$ How the Test Was Designed

- ▶ Model tested: ARBERT – trained on formal Arabic news (AFND)
- ▶ Same 500 samples used across all three versions
- ▶ Version 1: Original MSA → clean formal Arabic, no changes
- ▶ Version 2: Mild Dialect → light lexical swaps (negation, pronouns)
- ▶ Version 3: Heavy Dialect → aggressive substitutions across all key words
- ▶ Facts never changed – only the phrasing and register changed
- ▶ # # Any label flip = syntactic weakness, not factual confusion

— VIZ 1 —

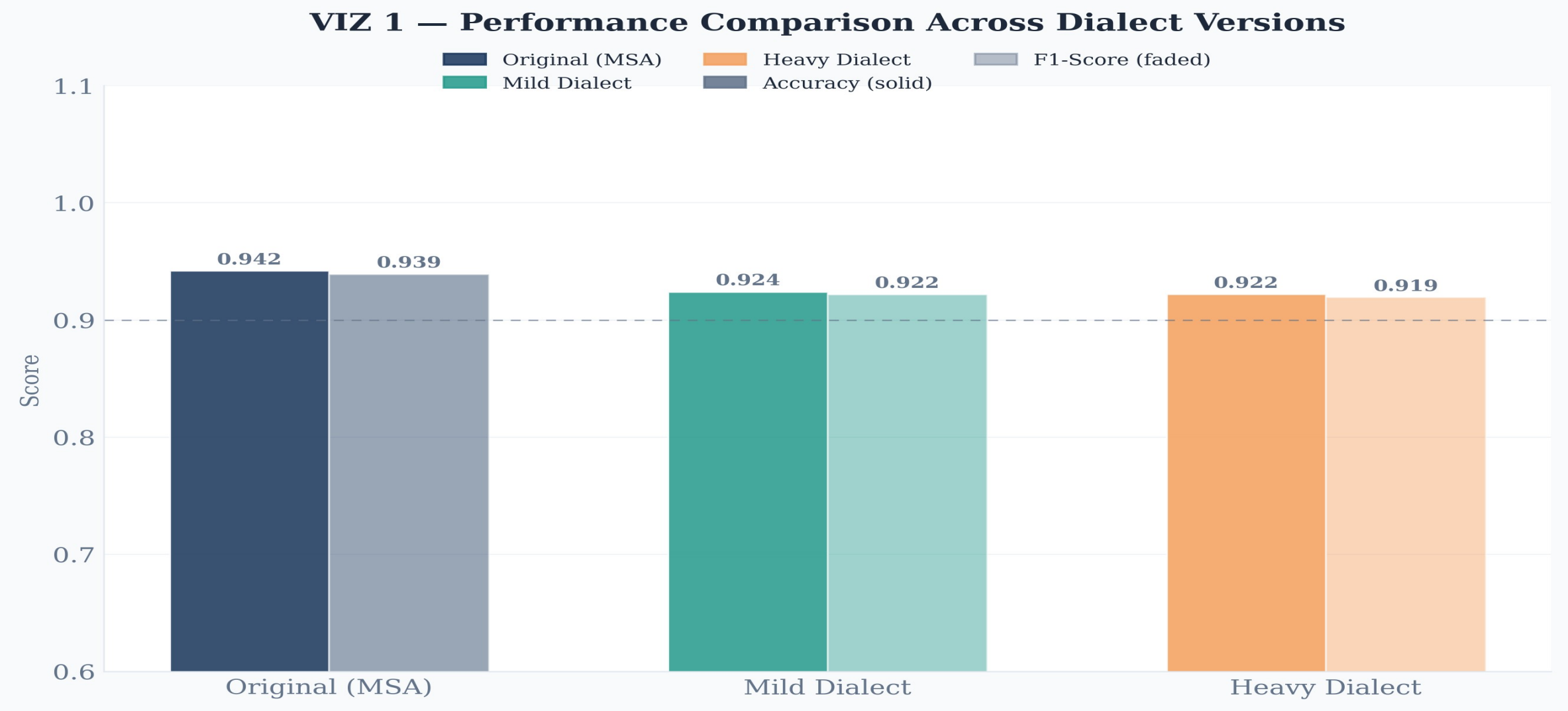
PERFORMANCE COMPARISON

Accuracy and F1 across all three dialect versions

\$ VIZ 1 – What the Bar Chart Shows

- ▶ Solid bars = Accuracy | Faded bars = F1-Score | Dashed line = 0.90 quality threshold
- ▶ Original MSA: Accuracy 0.942 | F1 0.939
- ▶ Mild Dialect: Accuracy 0.924 | F1 0.922 → still above 0.90
- ▶ Heavy Dialect: Accuracy 0.922 | F1 0.919 → barely above 0.90
- ▶ Accuracy and F1 drop almost equally – no class-specific bias introduced
- ▶ # # Consistent downward trend: MSA → Mild → Heavy confirms the Dialect Gap

VIZ 1 – Performance Comparison Across Dialect Versions



All three versions stay above 0.90 | Parallel drop in Accuracy and F1 – balanced, no class bias

— VIZ 2 —

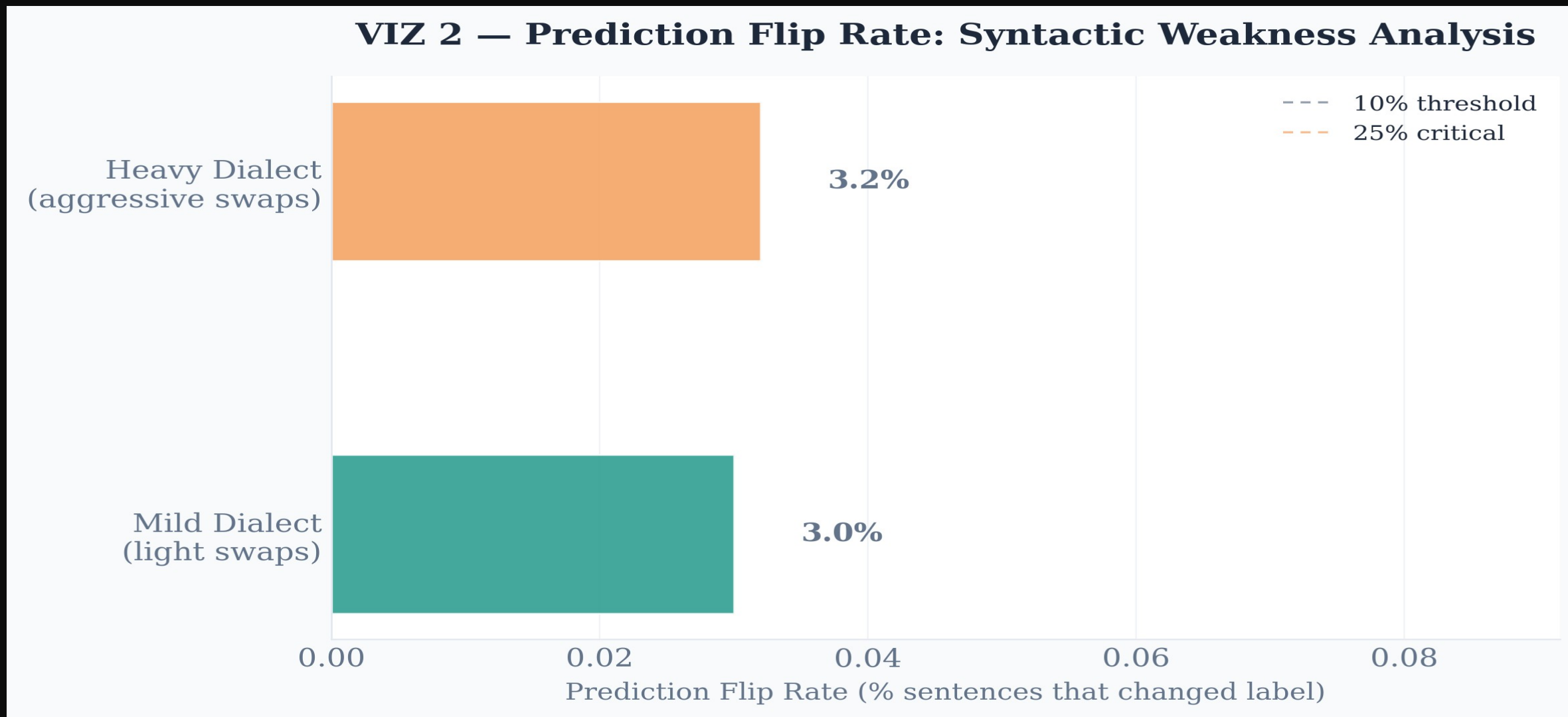
PREDICTION FLIP RATE

How often does dialect alone change the model's label?

\$ VIZ 2 – What the Flip Rate Tells Us

- ▶ Flip Rate = sentences where ARBERT changed its label after dialectal rewriting
 - ▶ Mild Dialect: 3.0% flip rate | Heavy Dialect: 3.2% flip rate
- ▶ Both below the 10% mild-concern threshold – label instability is low
- ▶ But flips exist: ~1 in 31 samples changed label with zero change to the facts
 - ▶ Tiny gap between Mild and Heavy → perturbation dictionary hit a coverage ceiling
 - ▶ # # Real social-media data would likely produce a steeper flip-rate curve

VIZ 2 – Prediction Flip Rate: Syntactic Weakness Analysis



Mild 3.0% | Heavy 3.2% – both below 10% threshold | Minimal gap shows diminishing returns

— VIZ 3 —

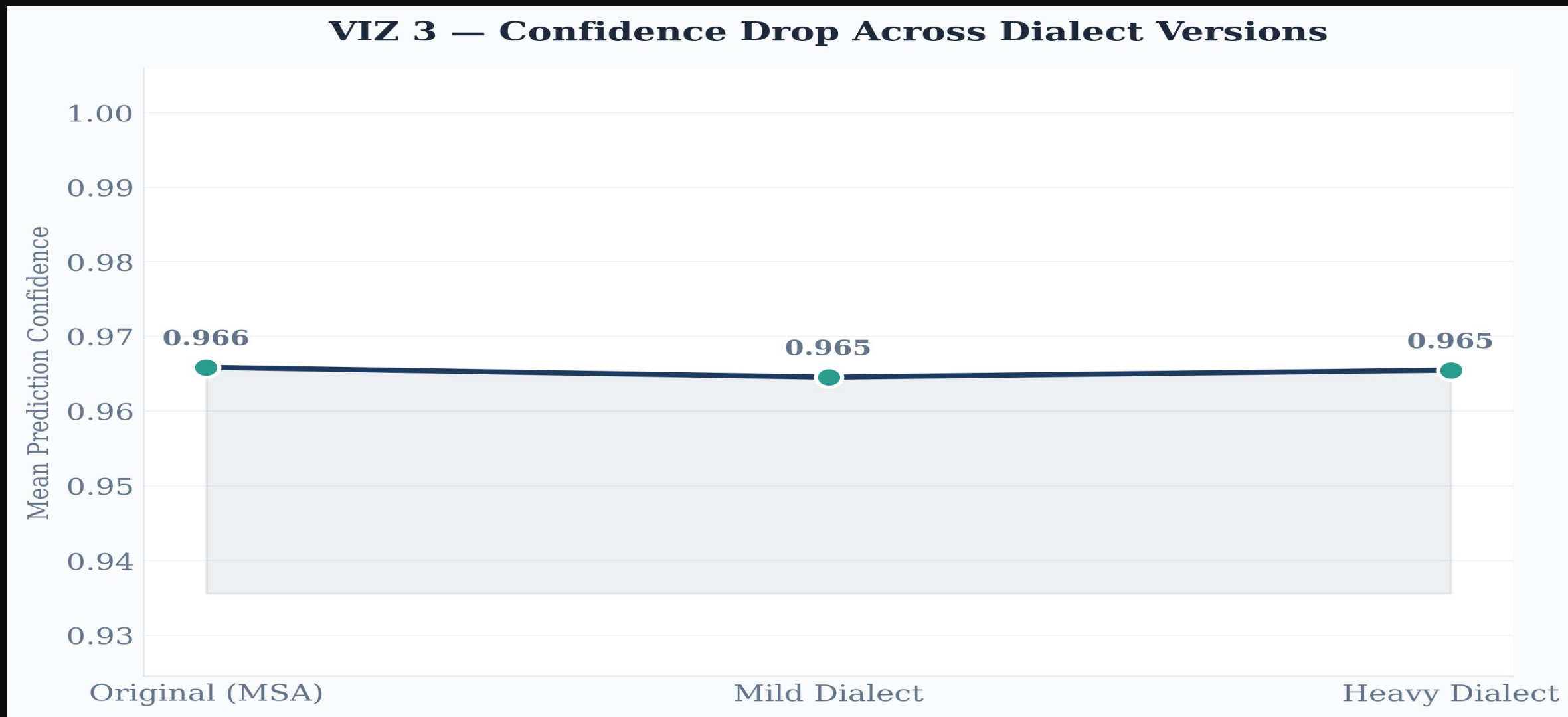
CONFIDENCE DROP

Does the model become less certain as dialect gets harder?

\$ VIZ 3 – The Overconfidence Problem

- ▶ Mean confidence across all three versions: 0.966 → 0.965 → 0.965
- ▶ Confidence stays almost completely flat – the model feels equally sure on all versions
- ▶ The critical gap: accuracy dropped, but confidence did NOT drop
- ▶ This is a calibration failure – the uncertainty signal cannot flag dialect-induced errors
- ▶ A system using confidence to trigger manual review would miss these mistakes entirely
- ▶ # # Dialect-aware confidence calibration is needed before production deployment

VIZ 3 – Confidence Drop Across Dialect Versions



Flat line at ~0.965 across all three versions | Accuracy drops but confidence stays – overconfidence on OOD input

— VIZ 4 —

CONFUSION MATRICES

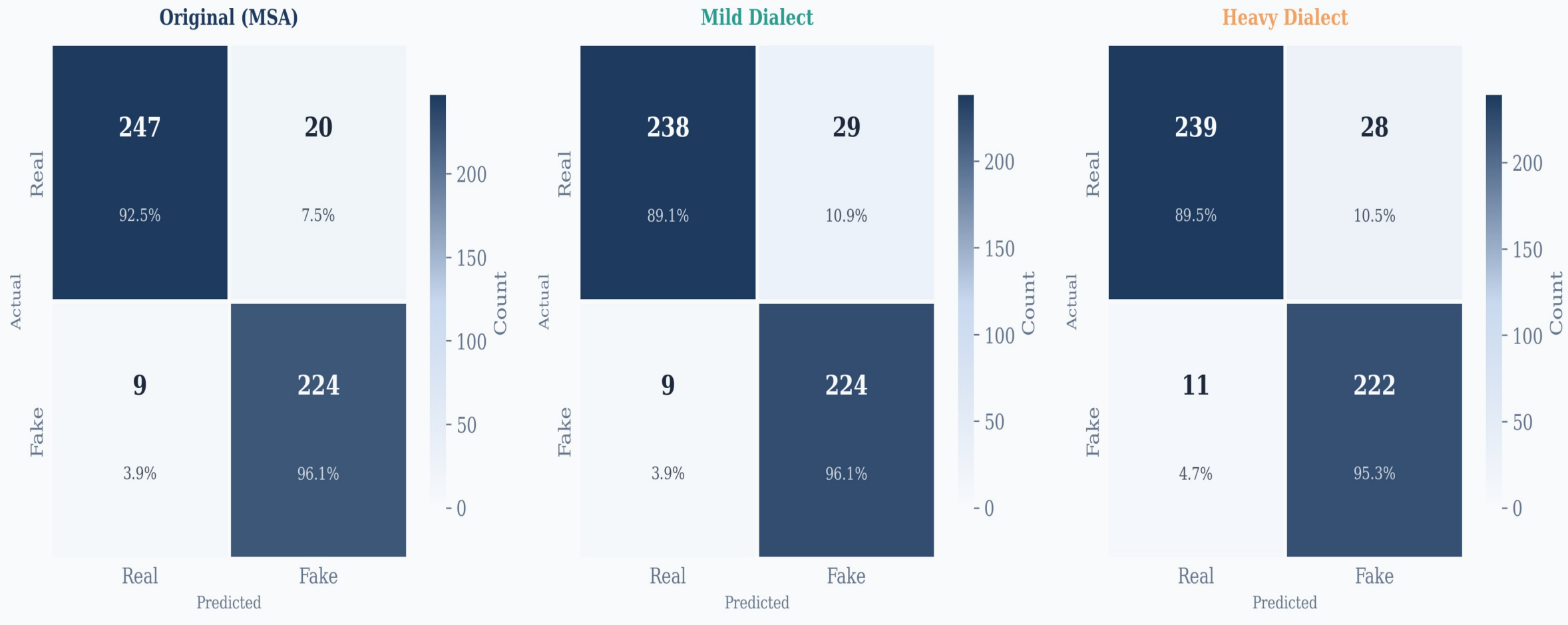
Where exactly do the errors grow as dialect increases?

\$ VIZ 4 – Reading the Three Matrices

- ▶ Fake detection stays stable: ~96% across all three versions
- ▶ Real news detection drops: 92.5% (MSA) → 89.1% (Mild) → 89.5% (Heavy)
- ▶ The Real→Fake cell is growing – dialectal phrasing makes the model suspicious of real news
 - ▶ Original MSA: 20 real→fake misses | Heavy Dialect: 28 real→fake misses
- ▶ 8 extra real samples falsely flagged as Fake – solely due to lexical substitution, no fact change
 - ▶ # # This is style-veracity bias: model judging HOW it's written, not WHAT it says

VIZ 4 – Confusion Matrices: Original / Mild / Heavy Dialect

VIZ 4 – Confusion Matrices: Original / Mild / Heavy Dialect



Fake row stays stable | Real→Fake cell grows: 20 → 29 → 28 misses | Dialect bias is real

— VIZ 5 —

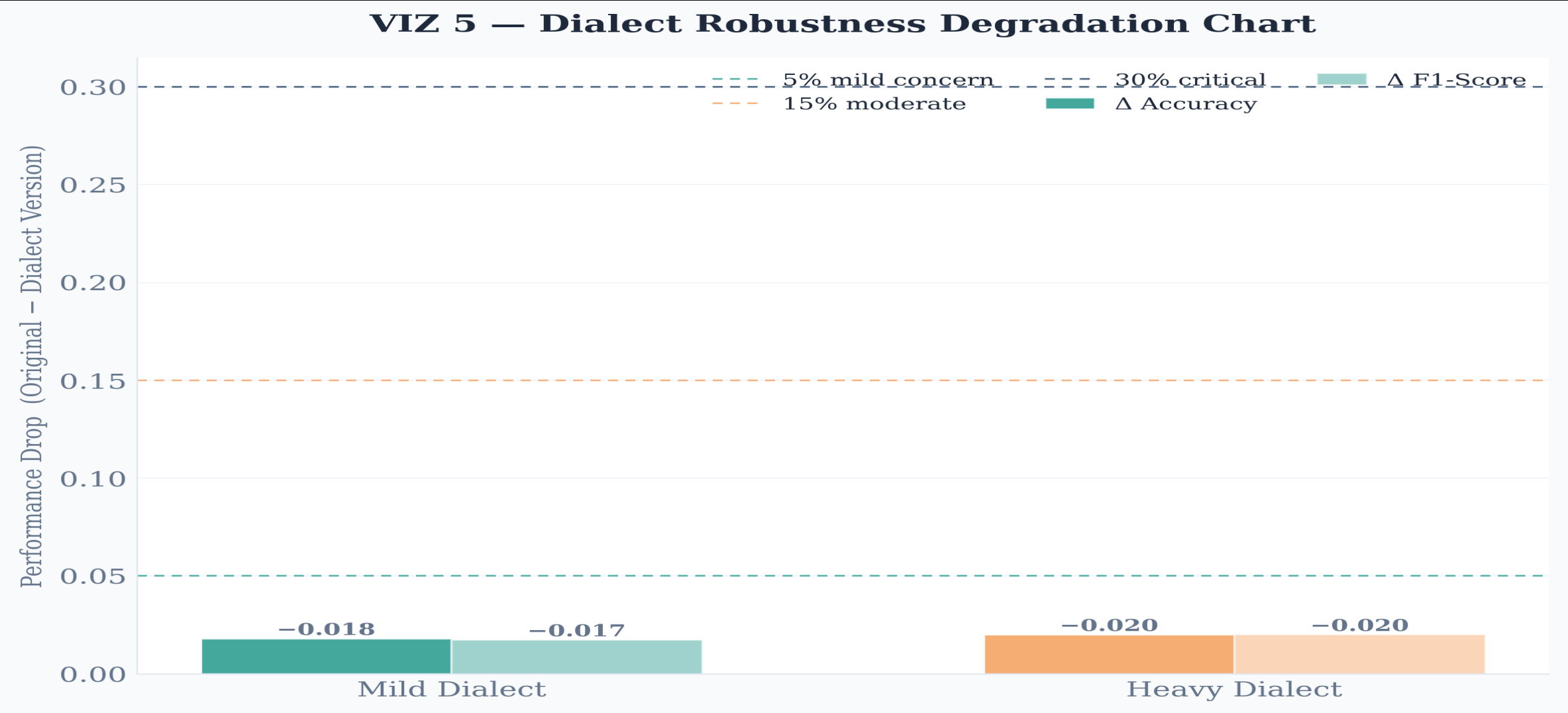
DIALECT ROBUSTNESS DEGRADATION

How big is the Dialect Gap – and where does it sit on the severity scale?

\$ VIZ 5 – Measuring the Dialect Gap

- ▶ Chart shows: Original minus Dialect version – how much performance was lost
 - ▶ Mild Dialect: $\Delta\text{Accuracy} = -0.018$ | $\Delta\text{F1} = -0.017$
 - ▶ Heavy Dialect: $\Delta\text{Accuracy} = -0.020$ | $\Delta\text{F1} = -0.020$
- ▶ Both in the LOW severity zone – well below the 5% mild-concern threshold
- ▶ Tiny gap between Mild and Heavy (only 0.002) – diminishing returns confirmed
 - ▶ # # ~2% drop shows robustness to rule-based perturbations – but real data could be worse

VIZ 5 – Dialect Robustness Degradation Chart



Both bars far below the 5% line | Mild -0.018 | Heavy -0.020 | Low severity confirmed

— CONCLUSION —

FINAL VERDICT

ARBERT is robust – but not fully safe for dialect-heavy deployment

\$ Key Takeaways

- Good news: ARBERT survives dialect pressure
 - ▶ All versions stay above 0.90 accuracy | Dialect Gap only ~2% |
Low severity zone
 - Hidden problem: overconfidence on out-of-domain input
 - ▶ Confidence stays flat while accuracy drops – the model doesn't know it's struggling
 - Structural bias found: Real news gets unfairly flagged as Fake under dialect
 - ▶ 8 extra false positives in Heavy version – style-veracity bias is measurable and real
 - ▶ # # Next step: test on actual social media Arabic data before declaring full robustness

[EOF]

Mohamed Shreif Abdelsattar | ID: 2405467 | Questions welcome

shreif@terminal:~# _